

Shrishti Barethiya

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Education

Ph.D. in Computational Chemistry

2022 – Expected 2027

Department of Chemistry, University of Massachusetts Amherst

Advisor: Dr. Jianhan Chen

Focus: Physics-informed ML for sequence-to-function prediction, Generative modeling of dynamic protein conformations, multi-scale simulation of intrinsically disordered proteins.

Integrated M.Sc. degree

2017 – 2022

School of Chemical Sciences, National Institute of Science Education and Research

Bhubaneswar, Odisha, India

Recipient of DISHA Scholarship, Department of Atomic Energy (Govt. of India)

Advisor: Dr. Upakarasamy Lourderaj

Focus: Machine learning for potential energy surface representation (potential energy surface) of small molecules.

Technical Skills

- **Machine Learning:** Deep learning (CNN, GNN, VAE, latent DDPM, Transformers, DiT), dimensionality reduction, clustering, traditional ML, PyTorch, JAX, TensorFlow, scikit-learn
- **Molecular Simulation:** All-atom (CHARMM, GROMACS, OpenMM) and coarse-grained modeling and simulation
- **Molecular Modeling Tools:** Rosetta, PyRosetta, MDAnalysis, Modeller, VMD, PyMOL
- **Programming:** Python (scientific computing, ML workflows), R, Bash, Git
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn, Origin
- **Computing Environments:** Linux (Ubuntu), Windows, MacOS, HPC clusters (SLURM)

Research Experience

Ph.D. Researcher, Computational Biophysics Lab, University of Massachusetts Amherst, MA

- Physics-informed Machine learning for Protein Sequence-to-Function Prediction
 - Developed CNNs and GCNs incorporating biophysics-based features for a robust protein variant effect prediction. Achieved significantly high performance for positional extrapolation.
 - Led collaborative project to integrate sequence, evolutionary, structural, and biophysical features to identify resistance-conferring mutations in Mycobacterium tuberculosis, resulting in a resistance forecast model.
- Generative Ensemble Modeling of Intrinsically Disordered Protein
 - Trained latent diffusion transformer models on a coarse-grained (CG) molecular dynamics dataset (~2.6M disordered protein conformations) to generate conformational ensembles for IDPs. Achieved a generative model that provides accurate secondary and global structure predictions.
 - Designed and trained a new latent diffusion transformer-based generative model using ESM embeddings and a folded protein structure database, aimed at improving accuracy beyond the CG disordered protein conformational dataset.
- HyRes-IDRome: A Large-Scale Simulation Dataset of Intrinsically Disordered Proteins from the Human Proteome and Other Species.
 - Developed a large-scale, open-source molecular simulation dataset comprising ~30,000 IDPs (including intrinsically disordered regions or IDRs) curated from the human IDRome and DisProt databases.
 - Extracted and curated key simulation and sequence-derived features, including radius of gyration, end-to-end distance, average residual helicity, helical segment annotations, and the Flory scaling exponent, as well as the GO annotation and taxonomic information to enable downstream computational analyses.

- Contributed 2,453 simulated IDR conformational ensembles to the Protein Ensemble Database (PED), a public benchmark resource for intrinsically disordered protein structure.
- Phosphorylation Dynamics of p53 via coarse-grained simulations
 - Investigated how phosphorylation in p53 modulates NTAD/DBD interactions governing autoinhibition of the tumor suppressor p53.

Master's thesis, Theoretical and Computational Chemistry Lab, National Institute of Science Education and Research Bhubaneswar, India

- Machine Learning for Chemical Dynamics Simulations
 - Applied neural networks and Gaussian process to represent potential energy surface of a three-dimensional system, enabling scalable modeling of molecular dynamics.
 - Compared neural network approaches with traditional fitting methods (non-linear least squares, cubic splines) for small-molecule potential energy surface representation.
 - Applied ANN models to predict outcomes of classical trajectory simulations, enabling sampling of states across higher energy barriers.

Publication

*: co-first author

- 6 A. M. Monzon, H. Ghafouri, X. A. Castro, S. Khezrian, M. C. Aspromonte, K. Singla, **S. Barethiya**, P. Bernado, M. Bonomi, J. Chen, T. Cordeiro, J. Cortés, S. De Castro, M. Feig, J. Forman-Kay, T. Head-Gordon, A. Holehouse, G. Janson, K. Lindorff-Larsen, Z. H. Liu, S. Longhi, M. Pajkos, P. Robustelli, S. Rodriguez, V. Schnapka, N. Sibille, M. Vendruscolo, S. Tosatto, PED in 2027: integrating experimental and predicted protein conformational ensembles at scale (submitted)
- 5 M. Tasmin, **S. Barethiya**, Y. Wang, L. Kang, A. Green, and J. Chen, Forecasting Antibiotic Resistance in *Mycobacterium tuberculosis* using Biophysics and Machine Learning *bioRxiv* (2026).
- 4 S. Li, **S. Barethiya**, J. Chen, HyRes: Accurate Physics-Based Simulation of Dynamic Protein Structures and Interactions in Complex Environments at Scale *bioRxiv* (2026).
- 3 **S. Barethiya**, J. Huang, C. Stumpf, X. Liu, H. Guan, and J. Chen, Overcoming extrapolation challenges of deep learning by incorporating physics in protein sequence-function modeling *PLOS Computational Biology* (2026).
- 2 **S. Barethiya***, S. Schultz*, Y. Zhang, and J. Chen, Coarse-Grained Simulations of Phosphorylation Regulation of p53 Autoinhibition. *Biochemistry* (2025).
- 1 E. B. Nordquist*, G. Zhang*, **S. Barethiya**, N. Ji, K. M. White, L. Han, Z. Jia, J. Shi, J. Cui, and J. Chen, Incorporating physics to overcome data scarcity in predictive modeling of protein function: a case study of BK channels. *PLOS Computational Biology* (2023)
- 0 L. Zheng*, **S. Barethiya***, E. B. Nordquist, and J. Chen, Machine learning generation of dynamic protein conformational ensembles. *Molecules* (2023)

Selected Talks and Posters

- Presented a talk in University of Massachusetts Amherst, ResearchFest 2026, *Physics- and language-model-guided generative modeling of dynamic protein ensembles*.
- Presented a poster at the Biophysical Society Annual Meeting 2025, *Overcoming Extrapolation Challenges of Deep Learning by Incorporating Physics in Protein Sequence-Function Modeling*.
- Presented a talk in University of Massachusetts Amherst, Chemistry-Biology Interface 2025, *Physics-informed machine learning for protein structure and function*.
- Presented poster University of Massachusetts Amherst, ResearchFest 2024, *Combining Physics and Machine Learning for Antibiotic Resistance Forecast*.